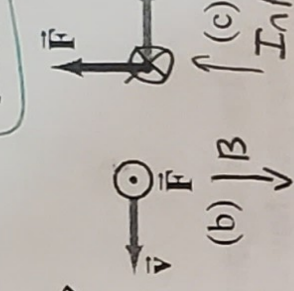
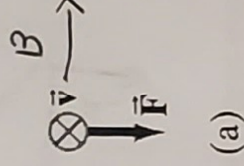
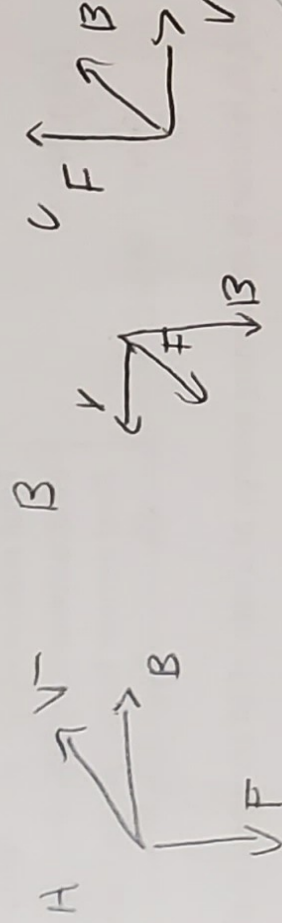


THINKING/INQUIRY

Determine the direction of $\vec{B}$ for each case in the shown diagram, where $\vec{F}$ represents the maximum magnetic force on a positively charged particle moving with velocity $\vec{v}$. Explain.

[3 marks]



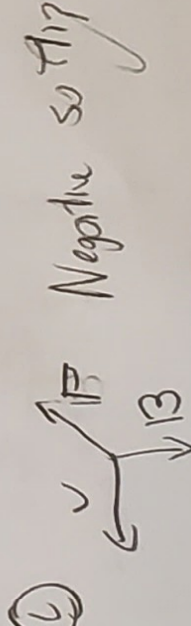
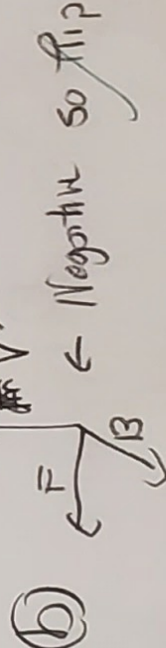
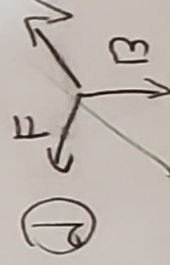
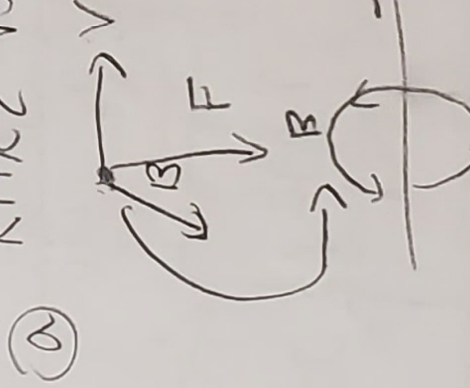
RHR 1

Final answer for a: east (Right)
Final answer for b: south (down)
Final answer for c: into the page

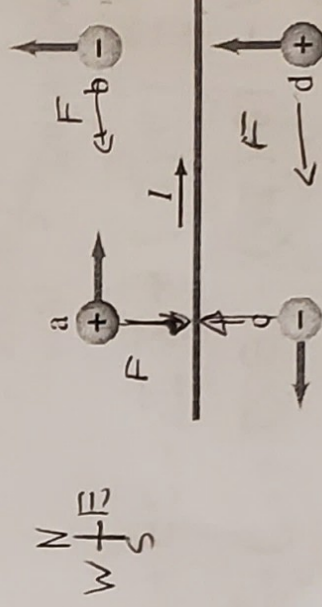
In the shown diagram, charged particles move in the vicinity of a current-carrying wire. For each charged particle, the arrow indicates the initial direction of motion of the particle, and the + or - indicates the sign of the charge. For each of the particles, indicate the direction of the magnetic force due to the magnetic field produced by the wire. Explain.

RHR 2 for direction of B

RHR for direction of Force



[4 marks]



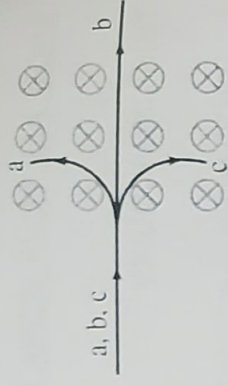
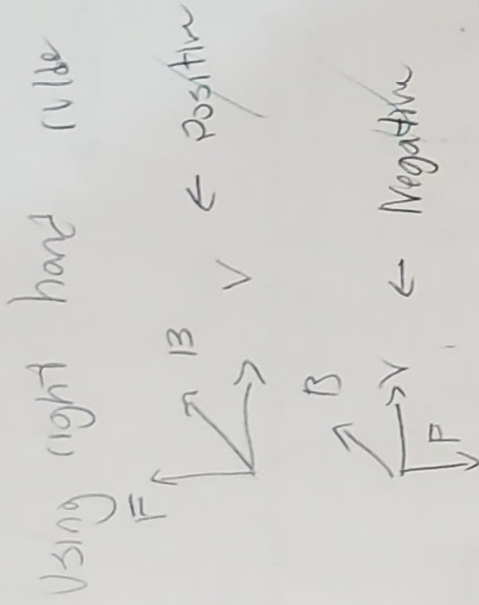
Final answer for a: South (down)
Final answer for b: West (left)
Final answer for c: North (up)
Final answer for d: West (left)

Name: _____

Date: _____

3. Three particles, a, b, and c, enter a magnetic field and follow paths as shown in the diagram. What can you say about the charge on each particle? Explain.

[13 marks]



Final answer for a: *positive* ✓
 Final answer for b: *neutral* ✓
 Final answer for c: *negative* ✓

4. In the circuit shown in the diagram, the $6.0\ \Omega$ resistor is consuming energy at a rate of $24\ \text{J/s}$ when the current through it flows as shown.

current

- a) The electric potential difference and electric field.
 b) What are the polarity and the voltage of the unknown battery, assuming it has negligible internal resistance? Show all your work.

$$P = IV = \frac{V^2}{R} = I^2 R$$

$$24 = I^2 R$$

$$\sqrt{\frac{24}{6}} = I$$

$$a) I = 0.233\ \text{A}$$

Thus current is $0.2\ \text{A}$ for resistors 3 and 4 because they are in series

b) Kirchhoff's law

$$V - 17(I_1) + 6I_3 - 25 + 3(I_4) - 13(I_2) = 0 \rightarrow 30\ \Omega$$

$$V - 17(I_1) - 30(I_{576}) - 13I_2 = 0$$

$$25 - 6I_3 - 30(I_{576}) - 3I_4 = 0$$

$$25 - 9(24) - 30(I_2) = 0 \quad I_2 = 0.233\ \text{A}$$

$$V - 30I_1 + 9(24) - 25 = 0$$

$$V - 30(-1.767) + 18 - 25 = 0 \quad I_1 + 2I_2 = I_2$$

$$V = -46\ \text{V}$$

$$= 0.233 - 2$$

$$I_1 = -1.767\ \text{A}$$

$$I_1 + I_3 = I_{576}$$

$$I_{576} = I_1 + I_2$$

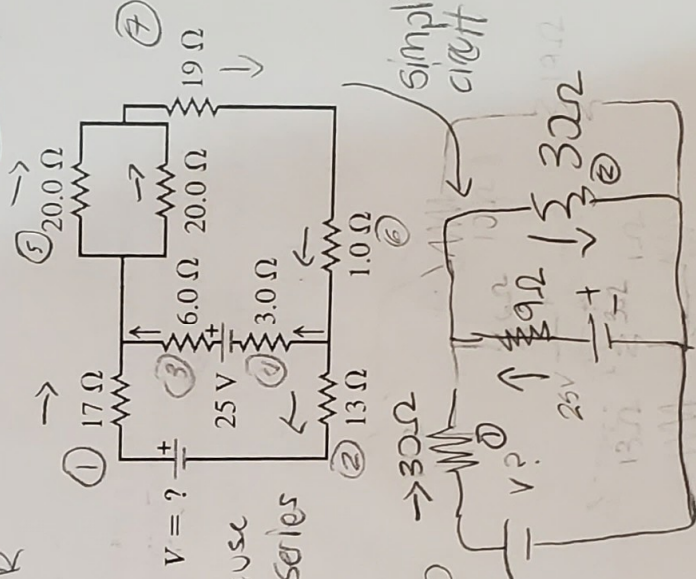
b) Polarity

$$- \frac{1}{+} \frac{46\ \text{V}}{+}$$

Polarity down

$$V = 46\ \text{V}$$

[5 marks]



Name: _____

Date: _____

5. Two long thin parallel wires 13.0 cm apart each carry 28 A current in the same direction. Determine the magnetic field vector at a point 10.0 cm from one wire and 6.0 cm from the other as shown in the diagram. Show all your work.

marks
[5 marks]

Find θ

$$B^2 = B_1^2 + B_2^2 - 2(B_1)(B_2)\cos\theta$$

$$0.809 = \cos\theta$$

$$26.34^\circ = \theta \quad \sin\theta = \frac{x}{10}$$

$$x = 4.43 \text{ cm} \quad \text{convert to meters}$$

$$x = 0.0443 \text{ m}$$

$$B_1 = \frac{\mu_0 I}{2\pi r} = \frac{4\pi \times 10^{-7} \cdot 28}{2\pi(0.06)} = 9.33 \times 10^{-5} \text{ T}$$

$$B_2 = \frac{4\pi \times 10^{-7} \cdot 28}{2\pi(0.1)} = 5.6 \times 10^{-5} \text{ T}$$

Vector addition

$$B_1 = 9.33 \times 10^{-5} \text{ T}$$

Calculations for the directions are included

$$R_2 = 3.81$$

$$x = B_2 + \cos 72^\circ B_1 = 8.16 \times 10^{-5} \text{ T}$$

$$y = \sin 72^\circ B_1 = 8.971 \times 10^{-5} \text{ T}$$

See Solutions!

$$9.52101 \times 10^{-5} \text{ T} = B_1 \text{ at } P$$

Next part of the problem

$$\text{Mag} = \sqrt{(8.97 \times 10^{-5})^2 + (8.16 \times 10^{-5})^2}$$

Name: _____

Date: _____

COMMUNICATION

1. Is current used up in a resistor? If yes, please explain. If not, what else gets used up?

[2 marks]
It travels through the resistor but does not drop, voltage drops when travelling through a resistor because work is done.

2. The Earth's "north pole" is really a magnetic south pole. Indicate how north and south magnetic poles were defined and how we can tell experimentally that the north pole is really a south magnetic pole.

[2 marks]
This is because the north of the compass points towards the south pole. They were defined as we the compass points North is North. But it is flipped since opposites attract.

3. If a negatively charged particle enters a region of uniform magnetic field which is perpendicular to the particle's velocity, will the kinetic energy of the particle increase, decrease, or stay the same? Explain your answer. (Neglect gravity and assume there is no electric field.)

[2 marks]
Stay the same since no work is done, only the direction changes. $W = \Delta KE$ $\vec{F} \perp \vec{v}$ $W = Fd \cos \theta$ And work change to $KE = \cos 90^\circ = 0$ $W = 0$

4. Can you set a resting electron into motion with a magnetic field? Explain.

[2 marks]
No, if you move the magnetic field then, but a velocity thus and force will be caused, a force $qVB \sin \theta = F$ $\vec{v} \perp \vec{B}$ so it is not possible unless you use EMF and a coil.

Communication marks are awarded throughout the test according to the shown rubric.

	Level 1	Level 2	Level 3	Level 4
Correct use of mathematical/physical symbols, labels, and conventions (e.g. units, proper component form for the values, positive direction given, etc...)	Few mathematical/physical conventions are used correctly. [1/2]	Some mathematical/physical conventions are used correctly. [1]	Most mathematical/physical conventions are used correctly. [1-1/2]	Consistently uses mathematical/physical conventions correctly. [2]

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

APPLICATION

1. Determine the magnitude of voltage V_1 and V_2 . Show all your work.

Loop one

Using loop rule Kirchhoff's

$$20V - (1A)(1\Omega) - (2A)(1\Omega) - V_2 - 2A(2\Omega)$$

$$7 - V_2 = 0$$

$$7V = V_2$$



$$-(6\Omega)(1A) = 0$$

Loop Two

$$-(1\Omega)(1A)$$

$$20V + (5\Omega)(1A) - V_1 - 6 = 0$$

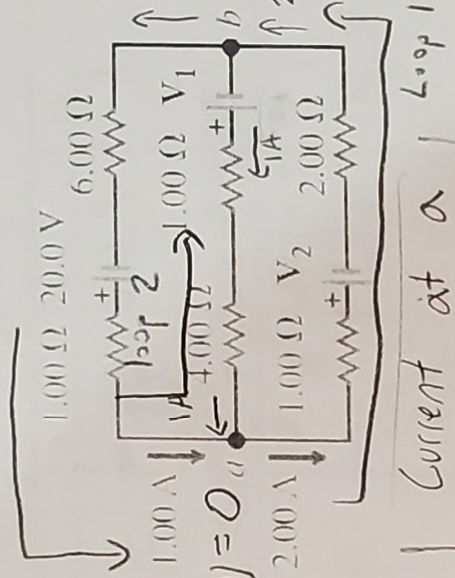
$$18 - V_1$$

$$= 0$$

$$18 = V_1$$



[6 marks]



Current at a

$$1 + I_A = 2$$

$$I_A = 1A$$

Current at B

$$2 = 1 + I_B$$

$$1A = I_B$$

Final answer: $V_2 = 7V$ $V_1 = 18V$

2. A 9.0-V battery whose internal resistance r is $0.50\ \Omega$ is connected in the circuit as shown in the diagram.

a) How much current is drawn from the battery?

b) What is the current in the $6.0\ \Omega$ resistor?

[4 marks]

Find equivalent resistance of circuit

$$R = \frac{1}{\frac{1}{8} + \frac{1}{4}} \leftarrow \text{smaller parallel}$$

$$R = 2.6667\ \Omega$$

$$2.6667 + 6\ \Omega = 8.6667\ \Omega \leftarrow \text{one half or parallel}$$

$$\frac{1}{8.6667} + \frac{1}{10} = R$$

combined parallel

$$4.64\ \Omega = R$$

series of entire circuit

$$4.64 + 0.5\ \Omega + 5\ \Omega = 10.14\ \Omega = R_{\text{equiv of whole circuit}}$$

$$V = IR$$

$$\frac{9}{10.14} = I$$

$0.887\text{ A} = I$ is the current drawn by circuit and battery

same voltage drop

$$9 - (5.5)(I) - V_2 = 0$$

$$9 - 5.5(0.887) - V_2 = 0$$

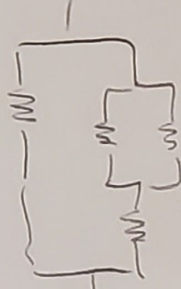
$$4.1197\text{ V} = V_2$$

$V = IR \leftarrow$ voltage through parallel section

is the same

$$\frac{4.1197}{8.662} = I$$

$$0.475\text{ A} = I$$



current through section is 0.475 and since it's in series then current through $6\ \Omega$ is 0.475 A as well.

KNOWLEDGE

For the multiple-choice questions below, circle the best answer and transfer the corresponding letter to the table provided.

Multiple Choice Answers:

1	2	3	4	5	6	7	8	9	10
C ✓	C ✓	B ✓	D ✓	C ✓	A ✓	B ✓	D ✓	B ✓	C ✓
11	12	13	14	15	16	17	18	19	20
B ✓	C ✓	A ✓	B ✓	A ✓	D ✓	B ✓	C ✓	D ✓	B ✓

- 1) How many electrons flow through a battery that delivers a current of 3.0 A for 12 s?

a. 4.8×10^{15}
 b. 6.4×10^{18}
 c. 2.2×10^{20}
 d. 3.6×10^{21}

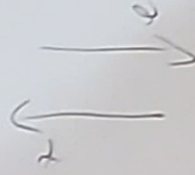
- 2) Complete the following statement: The unit kilowatt $\times$ hour measures

a. current.
 b. voltage.
 c. energy.
 d. power.

$$A \quad P \quad \frac{J}{s} \quad s = J \quad \checkmark$$

- 3) Which one of the following situations results in a conventional electric current that flows northward?

a. a beam of electrons moves northward
 b. a beam of electrons moves southward
 c. a beam of protons moves northward
 d. a beam of protons moves eastward



- 4) Which one of the following combinations of units is equivalent to the ohm?

a. W/A
 b. J/s
 c. V/C
 d. $J \times s/C^2$